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SOURCE: MEPC.219(63).pdf
amounts. The port may have special discharge requirements for food wastes and related garbage which may carry certain disease or pest organisms, dunnage, batteries, medicines, outdated pyrotechnics or unusually large, heavy, or odorous derelict fishing gear, etc.

2.8 Shipboard equipment for processing garbage

2.8.1 The choice of options 4 for garbage processing depends largely upon personnel limitations, generation rate, capacity, vessel configuration, voyage route and availability of port reception facilities . The type of equipment available for shipboard garbage handling includes incinerators, compactors, comminuters and their associated hardware.

2.9 Grinding or comminution

2.9.1 The discharge of comminuted food wastes may be permitted under regulations 4.1.1 and 6.1.1 of MARPOL Annex V whilst the ship is en route. Such comminuted or ground food waste must be capable of passing through a screen with openings no greater than 25 mm.

2.9.2 A wide variety of food waste grinders is available on the market and most modern ships' galleys have the equipment needed to produce a slurry of food particles and water that

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SOURCE: MEPC.219(63).pdf
2.9.2 A wide variety of food waste grinders is available on the market and most modern ships' galleys have the equipment needed to produce a slurry of food particles and water that washes easily through the required 25 mm screen. Output ranges from 10 to 250 litres per minute. The discharge from shipboard comminuters should be directed into an appropriately constructed holding tank when the vessel is operating within an area where discharge is prohibited.

2.9.3 Size reduction of certain other garbage items can be achieved by shredding or crushing and machines for carrying out this process are available for use on board ships.

2.9.4 Information on the development, advantages and use of comminuters for processing food waste aboard ships should be forwarded to the Organization for sharing between interested parties.

2.9.5 Outside special areas, ships operating primarily beyond three nautical miles from the nearest land are encouraged to install and use comminuters to grind food wastes to a particle size capable of passing through a screen with openings no larger than 25 mm.

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SOURCE: MEPC.219(63).pdf
the nearest land are encouraged to install and use comminuters to grind food wastes to a particle size capable of passing through a screen with openings no larger than 25 mm. Regulation 4 requires comminuting or grinding food wastes if the food wastes are to be discharged between three and 12 nautical miles from the nearest land. Although unprocessed food wastes may be discharged beyond 12 nautical miles, it is recommended

that comminuters be used as they hasten assimilation into the marine environment. Because food wastes comminuted with plastics cannot be discharged into the sea, all plastic materials need to be removed before food wastes are placed into a comminuter or grinder.

4 Reference may also be made to other technical guidance such as, ISO/CD21070 Ships and marine technology Marine environment protection Management and handling of shipboard garbage.

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SOURCE: MEPC.219(63).pdf
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2.9.6 When operating inside a special area, regulation 6 of MARPOL Annex V requires all food wastes to be comminuted or ground prior to discharge in to the sea. All discharges are to be as far as practicable and not less than 12 nautical miles from the nearest land or ice-shelf.

2.10 Compaction

Table 3 shows compaction options for various types of garbage.

Table 3 Compaction options for shipboard-generated garbage

Examples of garbage	Special handling by vessel personnel before compactIon	Compaction characteristics	Onboard storage space	Rate of alteration	Retainment of compacted form	Density of

compacted
form
Metal, food
and beverage
containers,
glass, small
wood pieces
None Very rapid Almost 100% High Minimum
Comminuted
plastics, fibre
and paper
board
Minor
reduce
material to
size for feed,
minimal
manual labour
Rapid Approximately
80%
Medium Minimum
Small metal
drums5,
uncomminuted
cargo packing,
large pieces of
wood
Moderate
longer manual

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SOURCE: MEPC.219(63).pdf
plastics, fibre
and paper
board
Minor
reduce
material to
size for feed,
minimal
manual labour
Rapid Approximately
80%
Medium Minimum
Small metal
drums5,
uncomminuted
cargo packing,
large pieces of
wood
Moderate
longer manual
labour time

required to
size material
for feed
Slow Approximately
50%
Relatively
low
Moderate
Uncomminuted
plastics
Major very
long manual
labour time to
size material
for feed;
usually
impractical
Very slow Less than 10% Very low Maximum
Bulky metal
cargo
containers,
thick metal
items